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Zander et al.

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(54) **BEAM MANAGEMENT DURING INITIAL ACCESS**

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(58) **Field of Classification Search**

CPC . H04B 7/0695; H04B 7/0404; H04W 74/006;
H04W 74/0833; H04L 5/0048

See application file for complete search history.

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(57) **ABSTRACT**

A method of operating a wireless communication device (102) configured to connect to a communications network (100) includes: monitoring for multiple downlink reference signals (411) transmitted by at least one access node of the communications network (100) using multiple downlink transmit beams (361-364), each one of the multiple downlink reference signals (411) being associated with a respective at least one of multiple random-access occasions (415-418); and accessing at least two random-access occasions (415, 418) of the multiple random-access occasions (415-418), the at least two random-access occasions (415, 418) being associated with at least two downlink reference signals (411) selected from the multiple downlink reference signals (411) based on said monitoring and establishing a multibeam operation between the wireless communication

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(51) **Int. Cl.**

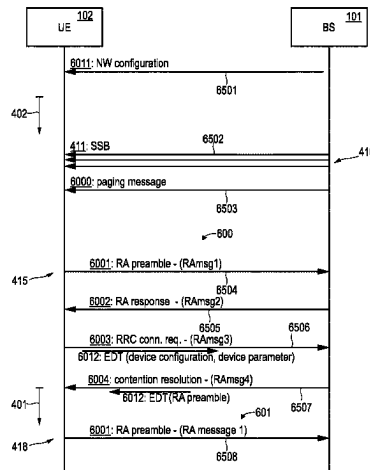
H04B 7/06 (2006.01)

H04W 74/00 (2009.01)

(Continued)

(52) **U.S. Cl.**

CPC **H04B 7/0695** (2013.01); **H04W 74/006** (2013.01); **H04W 74/0833** (2013.01)



device (102) and the communications network (100), wherein the multi-beam operation is established based on at least two downlink transmit beams (361,364) used for transmitting the selected at least two downlink reference signals (411).

20 Claims, 9 Drawing Sheets

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(51) Int. Cl.

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FIG. 1

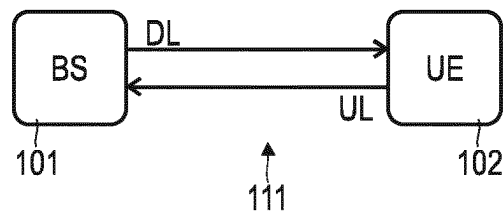


FIG. 2

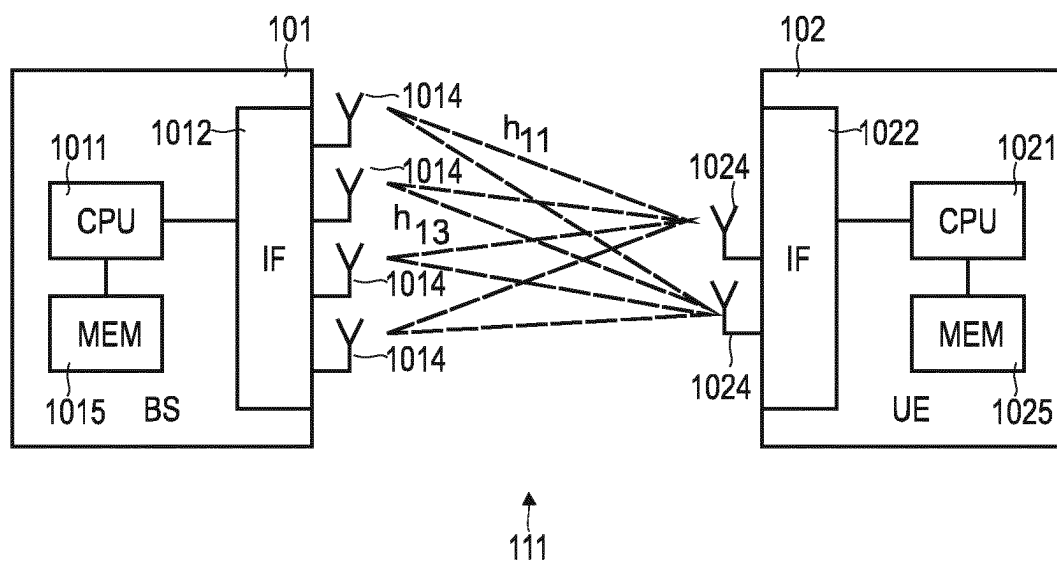


FIG. 3

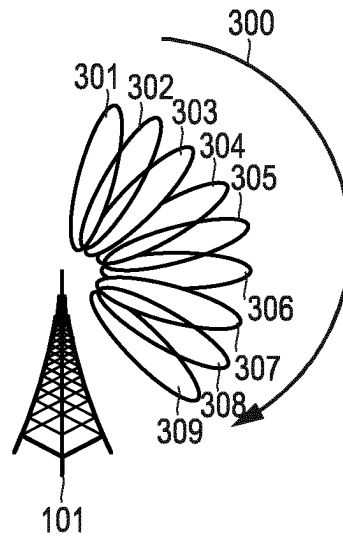


FIG. 4

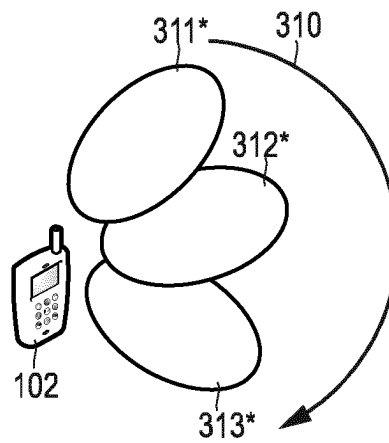


FIG. 5

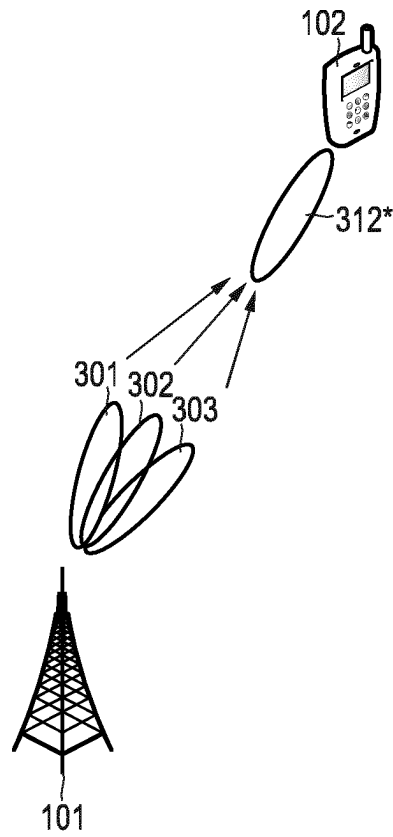


FIG. 6

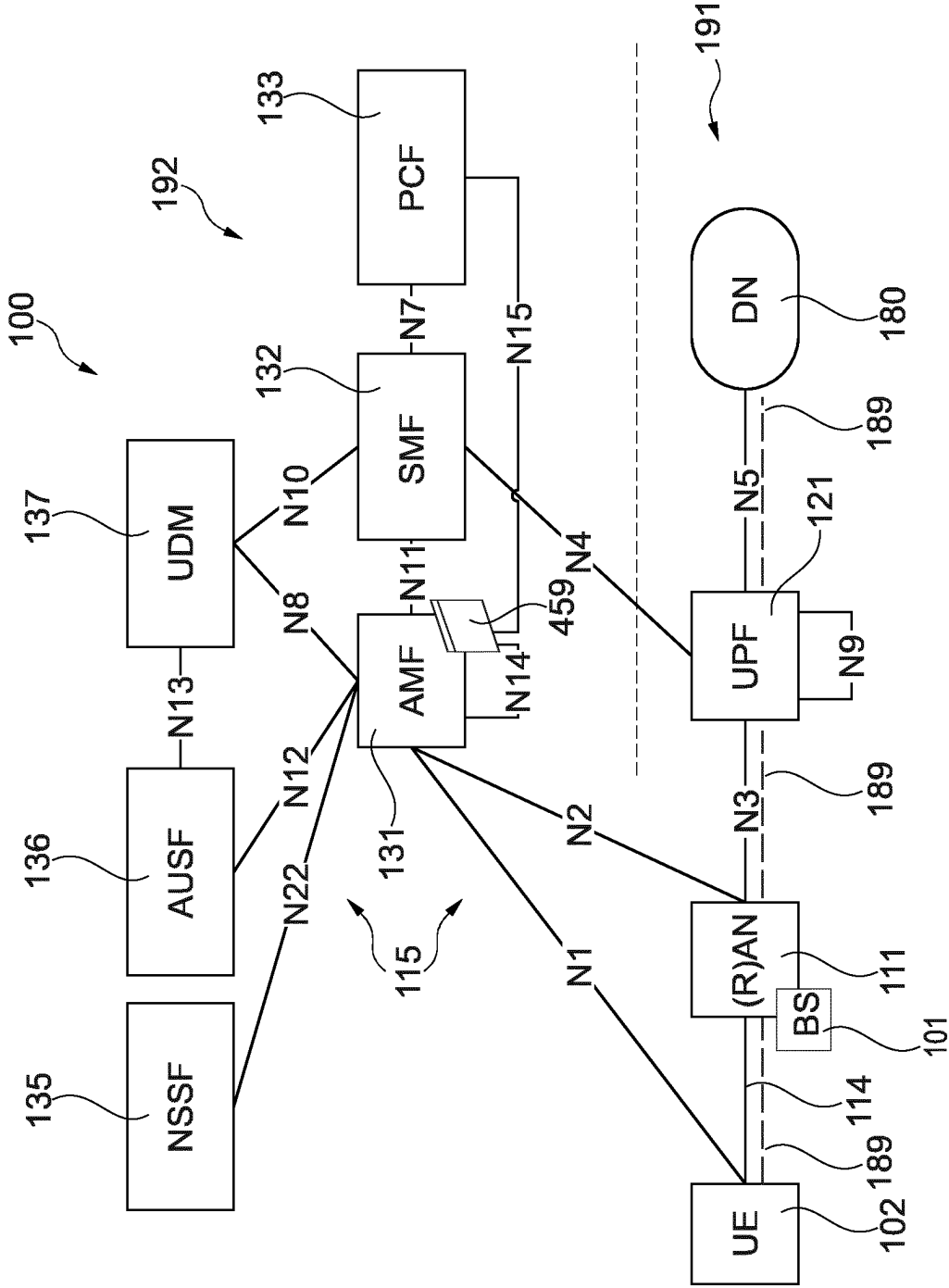


FIG. 7

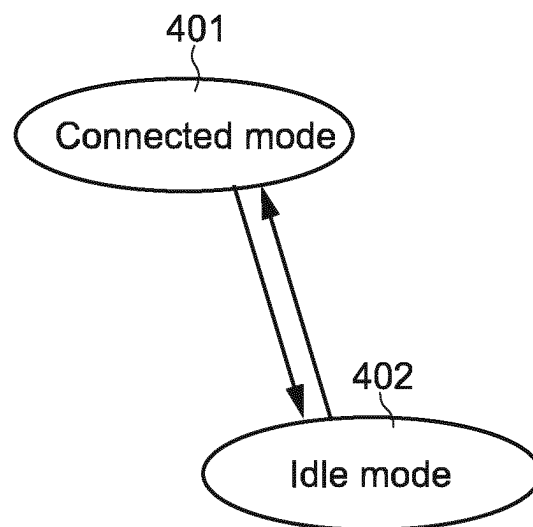


FIG. 8

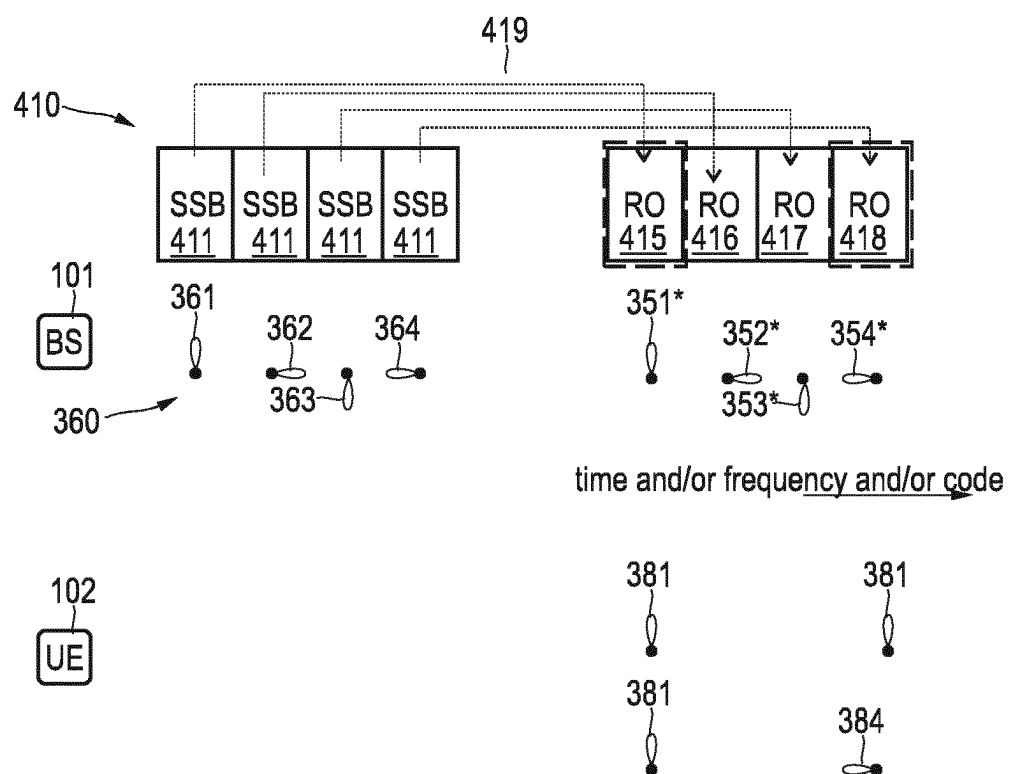


FIG. 9

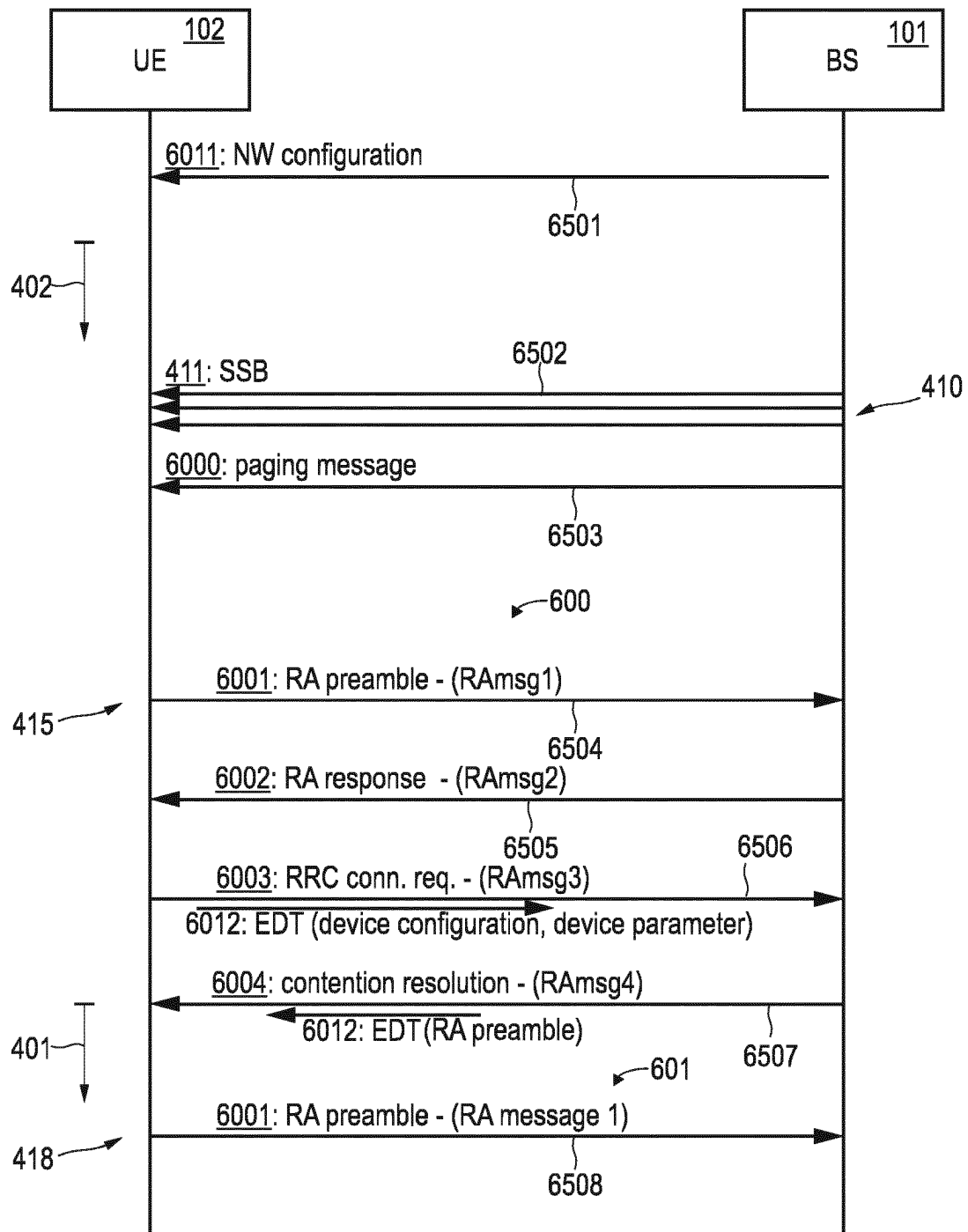


FIG. 10

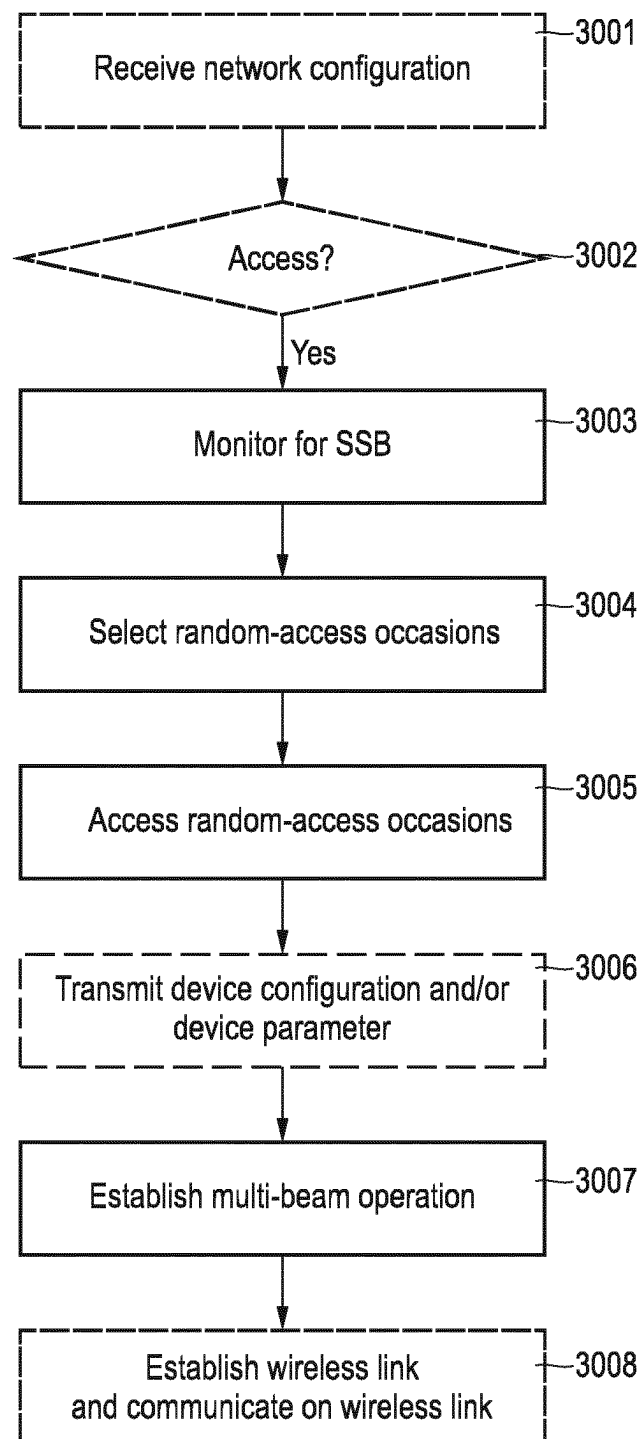
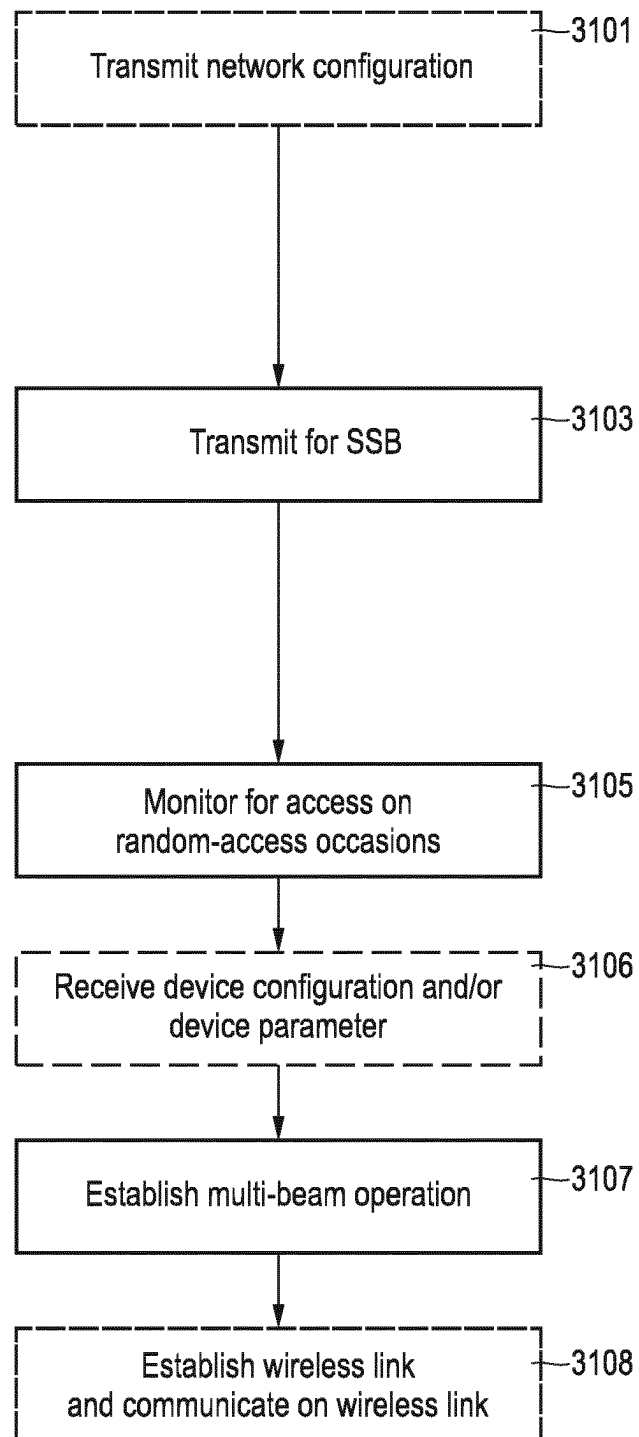


FIG. 11



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BEAM MANAGEMENT DURING INITIAL ACCESS

TECHNICAL FIELD

Various examples relate to beam management for a multi-beam operation in a communication system. More specifically, various examples relate to beam management during initial access of a wireless communication device to a cellular network.

BACKGROUND

Multiple-input multiple-output (MIMO) techniques are sometimes used to enhance reliability and/or throughput of communication on a wireless link. Here, the transmitter node and the receiver node both include multiple antennas that can be operated in a phase-coherent manner. Thereby, a signal can be transmitted redundantly (diversity multi-antenna mode) along multiple spatial data streams, or multiple signals can be transmitted on multiple spatial data streams (spatial multiplexing multi-antenna operational mode). Spatial data streams can be defined by focusing the transmission energy for transmitting (transmit beam, TX beam) and/or the receive sensitivity for receiving (receive beam, RX beam) to a particular spatial direction. Here, the process of identifying the appropriate beams is often referred to as beam establishment or beam management.

Beam management typically requires control signaling between the nodes of the MIMO communication system. For example, where a MIMO communication system is implemented by a base station of a cellular network and a wireless communication device attachable to the cellular network, initial access of the wireless communication device to the cellular network may impose a framework of control signaling that, according to reference implementations, makes it difficult to implement beam management at this early stage.

SUMMARY

Accordingly, there is a need for advanced techniques of beam management during initial access. This need is met by the features of the independent claims. The features of the dependent claims define embodiments.

A method of operating a UE is provided. The UE is configured to connect to a communications NW. The method includes monitoring for multiple downlink reference signals. The multiple downlink reference signals are transmitted by at least one access node of the communications network. The multiple downlink reference signals are transmitted using multiple downlink transmit beams. Each one of the multiple downlink reference signals is associated with a respective at least one of multiple random-access occasions. The method also includes accessing at least two RA occasions of the multiple RA occasions. The at least two RA occasions are associated with at least two downlink reference signals. The at least two downlink reference signals are selected from the multiple downlink reference signals. Said selecting is based on said monitoring. The method further includes establishing a multi-beam operation between the UE and the communications network. The multi-beam operation is established based on the at least two downlink transmit beams used for transmitting the selected at least two downlink reference signals.

A computer program or a computer-program product or a computer-readable storage medium includes program code.

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The program code can be executed by at least one processor. Executing the program code causes the at least one processor to perform a method of operating a UE. The UE is configured to connect to a communications NW. The method includes monitoring for multiple downlink reference signals. The multiple downlink reference signals are transmitted by at least one access node of the communications network. The multiple downlink reference signals are transmitted using multiple downlink transmit beams. Each one of the multiple downlink reference signals is associated with a respective at least one of multiple random-access occasions. The method also includes accessing at least two RA occasions of the multiple RA occasions. The at least two RA occasions are associated with at least two downlink reference signals. The at least two downlink reference signals are selected from the multiple downlink reference signals. Said selecting is based on said monitoring. The method further includes establishing a multi-beam operation between the UE and the communications network. The multi-beam operation is established based on the at least two downlink transmit beams used for transmitting the selected at least two downlink reference signals.

A wireless communication device is configured to connect to a communications network. The wireless communication device includes control circuitry configured to monitor for multiple downlink reference signals transmitted by at least one access node of the communications network using multiple downlink transmit beams, each one of the multiple downlink reference signals being associated with a respective at least one of multiple random-access occasions. The control circuitry is further configured to access at least two random-access occasions of the multiple random-access occasions, the at least two random-access occasions being associated with at least two downlink reference signals selected from the multiple downlink reference signals based on said monitoring. The control circuitry is further configured to establish a multi-beam operation between the wireless communication device and the communications network, wherein the multi-beam operation is established based on at least two downlink transmit beams used for transmitting the selected at least two downlink reference signals.

A method of operating at least one access node of a communications network is provided. The method includes transmitting multiple downlink reference signals. The multiple downlink reference signals are transmitted using multiple downlink transmit beams. Each one of the multiple downlink reference signals is associated with a respective at least one of multiple RA occasions. The method also includes monitoring an access of the UE to at least two RA occasions associated with at least two downlink reference signals of the multiple downlink reference signals. The method also includes establishing a multi-beam operation based on said monitoring of the access. The multi-beam operation is between the UE and the communications network. The multi-beam operation is established based on at least two downlink transmit beams of the multiple downlink transmit beams that are used for transmitting the at least two downlink reference signals.

A computer program or a computer-program product or a computer-readable storage medium includes program code. The program code can be executed by at least one processor. Executing the program code causes the at least one processor to perform a method of operating at least one access node of a communications network. The method includes transmitting multiple downlink reference signals. The multiple downlink reference signals are transmitted using multiple downlink transmit beams. Each one of the multiple down-

link reference signals is associated with a respective at least one of multiple RA occasions. The method also includes monitoring an access of the UE to at least two RA occasions associated with at least two downlink reference signals of the multiple downlink reference signals. The method also includes establishing a multi-beam operation based on said monitoring of the access. The multi-beam operation is between the EU and the communications network. The multi-beam operation is established based on at least two downlink transmit beams of the multiple downlink transmit beams that are used for transmitting the at least two downlink reference signals.

At least one access node of a communications network includes control circuitry. The control circuitry is configured to transmit multiple downlink reference signals using multiple downlink transmit beams, each one of the multiple downlink reference signals being associated with a respective at least one of multiple random-access occasions. The control circuitry is also configured to monitor an access of the wireless communication device to at least two random-access occasions associated with at least two downlink reference signals of the multiple downlink reference signals. Further, the control circuitry is further configured to, establish a multi-beam operation between the wireless communication device and the communications network based on said monitoring of the access, wherein the multi-beam operation is established based on at least two downlink transmit beams of the multiple downlink transmit beams used for transmitting the at least two downlink reference signals.

It is to be understood that the features mentioned above and those yet to be explained below may be used not only in the respective combinations indicated, but also in other combinations or in isolation without departing from the scope of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 schematically illustrates a communication system according to various examples.

FIG. 2 schematically illustrates MIMO operation of the communication system of FIG. 1 according to various examples.

FIG. 3 schematically illustrates a transmit beam sweep of a base station of the communication system according to various examples.

FIG. 4 schematically illustrates a receive beam sweep of a wireless communication device of the communication system according to various examples.

FIG. 5 schematically illustrates multi-beam operation of the wireless communication system according to various examples.

FIG. 6 schematically illustrates a cellular network according to various examples.

FIG. 7 schematically illustrates multiple connection modes in which the wireless communication device can be operated according to various examples.

FIG. 8 schematically illustrates multiple random-access occasions for initial access of the wireless communication device to the cellular network according to various examples.

FIG. 9 is a signaling diagram of communicating between the wireless communication device and the cellular network according to various examples.

FIG. 10 is a flowchart of a method according to various examples.

FIG. 11 is a flowchart of a method according to various examples.

DETAILED DESCRIPTION OF EMBODIMENTS

Some examples of the present disclosure generally provide for a plurality of circuits or other electrical devices. All references to the circuits and other electrical devices and the functionality provided by each are not intended to be limited to encompassing only what is illustrated and described herein. While particular labels may be assigned to the various circuits or other electrical devices disclosed, such labels are not intended to limit the scope of operation for the circuits and the other electrical devices. Such circuits and other electrical devices may be combined with each other and/or separated in any manner based on the particular type of electrical implementation that is desired. It is recognized that any circuit or other electrical device disclosed herein may include any number of microcontrollers, a graphics processor unit (GPU), integrated circuits, memory devices (e.g., FLASH, random access memory (RAM), read only memory (ROM), electrically programmable read only memory (EPROM), electrically erasable programmable read only memory (EEPROM), or other suitable variants thereof), and software which co-act with one another to perform operation(s) disclosed herein. In addition, any one or more of the electrical devices may be configured to execute a program code that is embodied in a non-transitory computer readable medium programmed to perform any number of the functions as disclosed.

In the following, embodiments of the invention will be described in detail with reference to the accompanying drawings. It is to be understood that the following description of embodiments is not to be taken in a limiting sense. The scope of the invention is not intended to be limited by the embodiments described hereinafter or by the drawings, which are taken to be illustrative only.

The drawings are to be regarded as being schematic representations and elements illustrated in the drawings are not necessarily shown to scale. Rather, the various elements are represented such that their function and general purpose become apparent to a person skilled in the art. Any connection or coupling between functional blocks, devices, components, or other physical or functional units shown in the drawings or described herein may also be implemented by an indirect connection or coupling. A coupling between components may also be established over a wireless connection. Functional blocks may be implemented in hardware, firmware, software, or a combination thereof.

Techniques are described that facilitate initial access of a UE to a communications network, e.g., to a cellular network. In particular, techniques are described that facilitate initial access using MIMO operation. In particular, the techniques described herein facilitate beam management during the initial access of a wireless communication device (UE) to a cellular network (NW). The techniques described herein can be used to reliably determine one or more downlink (DL) transmit (TX) beams at a base station (BS) of the cellular NW to be used for transmitting data to the UE. The one or more DL TX beams can then be used for subsequent communication of data, e.g., application data and/or higher-layer control data. Alternatively or additionally, using the techniques described herein, it is possible to determine one or more uplink (UL) receive (RX) beams at the BS. DL TX

beams and UL RX beams are so-called BS beams, because they are employed by the BS.

In some examples, it is also possible to determine UE beams, i.e., UL TX beams and/or DL RX beams. RX beams can be defined by a RX spatial filter. Alternatively or additionally, TX beams can be defined by a TX spatial filter. More generally speaking, a transmission using a spatial filter results in a certain beam or beam pattern.

Once a beam has been determined, this beam can be used for subsequent communication of data, e.g., application data and/or higher-layer control data.

According to various examples, it is possible for the UE to access multiple random-access (RA) occasions, the RA occasions (ROs) being associated with different BS beams. Based on this multi-RO access, it is then possible to establish a multi-beam operation. I.e., information conveyed by or determined based on the multi-RO access can be used to appropriately configure the multi-beam operation. For example, the BS can acquire information on multiple high-quality beam pairs between the BS and the UE already during initial access and at low delay. The overall control signaling overhead can be reduced. The capacity of the wireless link between the UE and the BS can be estimated. The multiple beam pairs determined from the multi-RO access can be used for the multi-beam operation.

For example, the UE may monitor for multiple DL reference signals (RSs). The DL RSs can be transmitted by the BS using multiple DL TX beams. Then, based on a comparison of RX properties of the multiple DL RSs—e.g., a RX amplitude—the UE can determine two or more strongest DL TX beams. The DL RSs can be associated with different ROs. The UE can accordingly select those ROs that are associated with the DL RSs having been transmitted using the two or more strongest DL TX beams. Then, the multi-beam operation—e.g., UL and/or DL multi-beam operation—can be established based on the two or more strongest DL TX beams and/or UL RX beams corresponding to those DL TX beams (e.g., having the same antenna weights). For example, the two or more strongest DL TX beams and/or the corresponding two or more UL RX beams may be used for the transmission in accordance with the multi-beam operation, e.g., for a subsequent data communication during a connected mode. It would also be possible to refine the two or more strongest DL TX beams and/or the corresponding two or more strongest UL RX beams: such beam refinement can include determining sub-beams, e.g., having a narrower beam width.

FIG. 1 schematically illustrates a communication system. The communication system includes two nodes **101**, **102** that are configured to communicate with each other via a wireless link **114**. In the example of FIG. 1, the node **101** is implemented by an access node, more specifically a BS, and the node **102** is implemented by a UE. The BS **101** can be part of a cellular NW (not shown in FIG. 1). As a general rule, the techniques described herein could be used for various types of communication systems, e.g., also for peer-to-peer communication, etc. For sake of simplicity, however, hereinafter, various techniques will be described in the context of a communication system that is implemented by a BS of a cellular NW and a UE.

Communication on the wireless link **114** can employ time-division duplex (TDD) and/or frequency-division duplex (FDD). Using TDD, communication in the DL and in the UL takes place at different points in time using the same frequency. Using FDD, communication in the DL and in the UL takes place at the same point in time, using different frequencies.

FIG. 2 illustrates details with respect to the BS **101**. The BS **101** includes control circuitry that is implemented by a processor **1011** and a non-volatile memory **1015**. The processor **1011** can load program code that is stored in the memory **1015**. The processor **1011** can then execute the program code. Executing the program code causes the processor to perform techniques as described herein, e.g.: communicating on a wireless link using MIMO; performing beamsweeps; transmitting signals such as RSs; scheduling signals for transmission on the wireless link; participating in initial access of the UE **102**; monitoring for the UE **102** accessing ROs; participating in establishment of a MIMO operation; participating in data transmission in accordance with the MIMO operation; etc.

FIG. 2 also illustrates details with respect to the UE **102**. The UE **102** includes control circuitry that is implemented by a processor **1021** and a nonvolatile memory **1025**. The processor **1021** can load program code that is stored in the memory **1025**. The processor can execute the program code. Executing the program code causes the processor to perform techniques as described herein, e.g.: communicating on a wireless link using MIMO and/or FDD; performing beamsweeps; participating in an initial access of the UE **102** to the cellular NW of the BS **101**; monitoring for RSs, i.e., attempting to receive RSs; selecting one or more ROs; participating in establishment of a MIMO operation; participating in data transmission in accordance with the MIMO operation; etc.

FIG. 2 also illustrates details with respect to communication between the BS **101** and the UE **102** on the wireless link **114**. The BS **101** includes an interface **1012** that can access and control multiple antennas **1014**. Likewise, the UE **102** includes an interface **1022** that can access and control multiple antennas **1024**.

While the scenario of FIG. 2 illustrates the antennas **1014** being coupled to the BS **101**, as a general rule, it would be possible to employ transmit-receive points (TRPs) that are spaced apart from the BS.

The interfaces **1012**, **1022** can each include one or more TX chains and/or more receiver chains. For instance, such RX chains can include low noise amplifiers, analogue to digital converters, mixers, etc. Analogue and/or digital beamforming would be possible.

Thereby, phase-coherent transmitting and/or receiving (communicating) can be implemented across the multiple antennas **1014**, **1024**. Thereby, the BS **101** and the UE **102** implement a MIMO communication system.

As a general rule, the receiver of the MIMO communication system receives a signal y that is obtained from an input signal x multiplied by the transmission matrix H . FIG. 2 includes two example labels for the components h_{11} and h_{13} of the transmission matrix H .

The transmission matrix H defines the channel impulse response of the wireless link **114**. The rank of the transmission matrix corresponds to the number of linearly independent rows or columns and, as such, indicates how many independent data streams can be used simultaneously; this is sometimes referred to as the number of layers. The rank can be set in different MIMO transmission modes. For MIMO transmission modes, the amplitude and/or phase (antenna weights) of each one of the antennas **1014**, **1024** is appropriately controlled by the interfaces **1012**, **1022**.

For instance, a diversity MIMO transmission mode relies on transmitting the same data redundantly using multiple TX antennas. Thus, multiple data streams carry the same data.

Thereby, the signal-to-noise ratio can be increased. The redundant signals can be generated using different encoding, e.g., Alamouti encoding.

Another MIMO transmission mode is spatial multiplexing. Spatial multiplexing allows to increase the data rate: The data is divided into different data streams and these different data streams can be transmitted contemporaneously over the wireless link **114**.

The diversity MIMO transmission mode and the spatial multiplexing multi-antenna transmission mode can be described as using multiple beams, the beams defining the spatial data streams. These modes are, therefore, also referred to as multi-beam operation. By using a beam, the direction of the wavefront of signals transmitted by a transmitter of the communication system is controlled. Energy is focused into a respective direction, by phase-coherent superposition of the individual signals originating from each antenna **1014**, **1024**. Thereby, the spatial data stream can be directed. The spatial data streams transmitted on multiple beams can be independent, resulting in spatial multiplexing multi-antenna transmission; or dependent on each other, e.g., redundant, resulting in diversity MIMO transmission.

As a general rule, alternatively or additionally to such TX beams, it is possible to employ RX beams.

The concept of beams can be used in so-called beam-sweeps. Details with respect to beamsweeping are explained next in connection with FIG. 3 and FIG. 4.

FIG. 3 illustrates DL TX beams **301-309** used by the BS **101**. In some examples, the BS **101** can employ a beam-sweep **300**. Here, the BS **101** activates the beams **301-309** on different resources (e.g., different time-frequency resources, and/or using orthogonal codes) such that the UE **102** can monitor for respective signals transmitted on the DL TX beams **301-309**. As a general rule, monitoring for signals can pertain to attempting to receive signals. This can include blind decoding, etc. The UE **102** may then select the best DL TX beam **301-309**—e.g., based on a RX property of the received signal, such as amplitude and/or phase and/or angle-of-arrival, etc.—and provide a respective indication to the BS **101**. As a general rule, the best DL TX beam **301-309** could be the strongest beam. Also, other quality metrics may be taken into account, e.g., signal-to-noise. Then, subsequent data can be communicated on the selected DL TX beam **301-309**. Such beam management can, in particular, be facilitated by using RSs, i.e., signals having a well-defined transmit property such as sequence, amplitude, phase, and/or precoding, etc.. RSs are sometimes also referred to as pilot signals.

It would be possible that such DL TX beamsweep **300** implemented by the BS **101** is coordinated with a DL RX beamsweep **310** implemented by the UE **102**. A DL RX beamsweep **310** is illustrated in FIG. 4. The DL RX beamsweep **310** includes multiple DL RX beams **311-313*** (for sake of simplicity, throughout this text, RX beams are denoted with “*”). The DL RX beamsweep **310** does not necessarily have to be coordinated with a DL TX beamsweep **300**.

FIG. 5 schematically illustrates aspects with respect to the diversity MIMO transmission mode. Here, the multiple transmit antennas are used to define multiple spatial data streams that carry the same data; this can also be viewed as a coherent combination of multiple DL TX beams **301-303** with optimized phase shifts.

FIG. 6 schematically illustrates a cellular NW **100**. The example of FIG. 6 illustrates the cellular NW **100** according to the 3GPP NR/5G architecture. Details of the 3GPP 5G

architecture are described in 3GPP TS 23.501, version 15.3.0 (2017-09). While FIG. 1 and further parts of the following description illustrate techniques in the 3GPP 5G framework of a cellular NW, similar techniques may be readily applied to other communication protocols. Examples include 3GPP LTE 4G—e.g., in the MTC or NB-IOT framework—and even non-cellular wireless systems, e.g., an IEEE Wi-Fi technology.

In the scenario of FIG. 1, a UE **102** is connectable to the cellular NW **100** via a data connection. For example, the UE **102** may be one of the following: a cellular phone; a smart phone; an IOT device; an MTC device; a sensor; an actuator; etc.

The UE **102** is connectable to a core NW (CN) **115** of the cellular NW **100** via a RAN **111**, typically formed by one or more BSs **101** (only a single BS **101** is illustrated in FIG. 1 for sake of simplicity). A wireless link **114** is established between the RAN **111**—specifically between one or more of the BSs **112** of the RAN **111**—and the UE **102**.

The wireless link **114** implements a time-frequency resource grid. Typically, OFDM is used: here, a carrier includes multiple subcarriers. The subcarriers (in frequency domain) and the symbols (in time domain) then define time-frequency resource elements of the time-frequency resource grid. Thereby, a protocol time base is defined, e.g., by the duration of frames and subframes including multiple symbols and the start and stop positions of the frames and subframes. Different time-frequency resource elements can be allocated to different logical channels or RSs of the wireless link **114**.

Examples include: Physical DL Shared Channel (PDSCH); Physical DL Control Channel (PDCCH); Physical UL Shared Channel (PUSCH); Physical UL Control Channel (PUCCH); channels for random access; etc.. For FDD, the PUCCH and PUSCH are communicated on the wireless link **114** in the UL frequency band **601** and the PDCCH and the PDSCH are in the DL frequency band **602** (hence, the names “UL frequency band” and “DL frequency band”).

The CN **115** includes a user plane (UP) **191** and a control plane (CP) **192**. Application data—e.g., of a data service—is typically routed via the UP **191**. For this, there is provided a UP function (UPF) **121**. The UPF **121** may implement router functionality. Application data may pass through one or more UPFs **121**. In the scenario of FIG. 1, the UPF **121** acts as a gateway towards a data NW (DN) **180**, e.g., the Internet or a Local Area NW. Application data can be communicated between the UE **102** and one or more servers **181** of the data NW **180**. The server **181** can execute an application that provides a service associated with the application data.

The cellular NW **100** also includes a mobility-control node, here implemented by an Access and Mobility Management Function (AMF) **131**. The cellular NW **100** also includes a session-control node, here implemented by a Session Management Function (SMF) **132**. The cellular NW **100** further includes a Policy Control Function (PCF) **133**; a NW Slice Selection Function (NSSF) **135**; an Authentication Server Function (AUSF) **136**; and a Unified Data Management (UDM) **137**.

FIG. 6 also illustrates the protocol reference points N1-N22 between these nodes.

The AMF **131** provides one or more of the following functionalities: connection management sometimes also referred to as registration management; NAS termination for communication between the CN **115** and the UE **102**; connection management; reachability management; mobil-

ity management; connection authentication; and connection authorization. After UE registration to the NW, the AMF 131 creates a UE context 459 and keeps this UE context, at least as long as the UE 102 is registered to the cellular NW 100. The UE context 459 can hold one or more identities of the UE 102 (UE ID), e.g., temporary identities.

The SMF 132 supports a data connection 189. The data connection 189 is established by the SMF 132 if the respective UE 102 operates in a connected connection mode. The data connection 189 is established on the wireless link 114. The wireless link 114 can pertain to the underlying physical medium and the data connection 189 can include a set of logical channels, scheduling rules, etc. The data connection 189 may include one or more data flows or bearers such as a dedicated data flow/bearer or a default data flow/bearer. The state of the data connection is also defined on the RRC layer, e.g., generally Layer 3 of the OSI model. The SMF 132 provides one or more of the following functionalities: session management including session establishment, modify and release, including data flow/bearer set up of UP data flows/bearers between the RAN 111 and the UPF 121; selection and control of UPFs; configuring of traffic steering; etc.

FIG. 7 illustrates aspects with respect to different connection modes 401-402 in which the UE 102 can operate according to various examples. When operating in the connected connection mode 401, the data connection 189 between the UE 102 and the cellular NW 100 is established. Application data can be communicated to and from the UE 102. A transition into the connected connection mode 401 is referred to as initial access.

When operating the UE 102 in the idle connection mode 402, the data connection 189 is released. It would be possible that the AMF 131 maintains the UE context 459. The UE 102 can use a discontinuous reception cycle (DRX) cycle that alternately switches the interface 1022 between an inactive state and an active state. The interface 1022—when operating in the inactive state—may not be able to RX any signals. E.g., an amplifier or an analogue to digital converter or parts of the digital front end may be switched off. On the other hand, the interface 1022—when operating in the active state—may be able to RX signals, e.g., paging signals.

To transition from the idle connection mode 402 to the connected connection mode 401, a RA procedure may be performed. The RA procedure includes transmitting a RA preamble at a RO. Details with respect to the ROs are illustrated in FIG. 8.

FIG. 8 schematically illustrates aspects with respect to synchronization signal blocks (SSBs). FIG. 8 illustrates that the BS 101 transmits multiple SSBs 411 at different resources, e.g., time-frequency and/or code resources.

The SSBs 411 can facilitate a cell search for the UE 102 during initial access. An SSB 411 can, e.g., include a primary synchronization signal (SS) and a secondary SS. The primary SS and the secondary SS implement RSs. The SSBs 411 can include a physical broadcast channel (PBCH). The PBCH carries a master information block and uses a specific modulation, typically quadrature phase shift keying, for reliable reception.

Within an SSB 411, the primary SS and the secondary SS are for the UE 102 to lock in to the correct frequency and time. There is also some identity of the transmitting BS 101 encoded into the combined SS. When in sync, the UE 102 can extract information about the associated RA channel allocation from the PBCH.

The SSBs 411 can be repeatedly or periodically transmitted by the BS 101, e.g., on predefined time-frequency resources of the time-frequency resource grid.

It is possible that a burst 410 of SSBs 411 is transmitted, employing a beamsweep 360 including the DL TX beams 361-364. Thereby, spatial coverage can be increased. Typically, the burst may have a duration of, e.g., up to 5 ms. The periodicity with which the bursts are transmitted may be 20 ms to 160 ms.

The SSBs 411 transmitted on different DL TX beams 361-364 can have a different SSB time index. Thus, different SSBs 411—i.e., having different SSB time indices—are associated with different ROs 415-418.

FIG. 8 also illustrates aspects with respect to the ROs 415-418. As illustrated in FIG. 8, there is an association 419 between SSBs 411 of the burst 410 having different SSB time indices, and multiple ROs 415-418. The different ROs 415-418 are arranged at different times, frequencies or employ different—e.g., orthogonal—codes for the RA preamble.

The association 419 between the SSBs 411 and the ROs 415-418 is typically implemented by pre-defining a number of SSB time indices per RO. This number can be larger than one which corresponds to multiple SSBs being associated with a single RO. The number can also be smaller than one which corresponds to a single SSB corresponding to multiple ROs. The SSBs (e.g., in ascending order of the time index) are then associated with the ROs, firstly, in frequency domain; secondly, in time domain within a slot; and thirdly in time domain between RA slots.

By selecting the appropriate RO 415-418, the UE 102 can indicate—according to reference implementations—which DL TX beam 361-364 was best. This enables to establish an appropriate beam pair during initial access.

The general framework of beam management during initial access has been described above. This framework of beam management during initial access can be enhanced by inventive concepts pertaining to multi-RO access. Such examples will be described next.

In the example of FIG. 8, the UE 102 accesses two ROs 415, 418, to establish multiple beam pairs. As a general rule, in the various examples described herein, it is possible that the UE 102 accesses two or more ROs. Then, multi-beam operation can be established on the wireless link 114 using the BS beams associated with the selected ROs: for DL transmission, this would include the DL TX beams 361, 364; and for UL transmission this would include the UL RX beams 351* and 354*. The UL RX beam 351* corresponds to the DL TX beam 361 in that it has a similar spatial characteristic and/or assumed beam correspondence. For example, the same antenna weights may be used for the BS beams 351* and 361.

FIG. 8 also illustrates aspects with respect to beam management at the UE 102 for UL TX beams to be used by the UE 102 to access the multiple ROs. There are various options available for implementing the access to the multiple ROs at the UE 102. Two of these options are illustrated in FIG. 8: in a first option (option I), it would be possible that the UE 102 uses the same UL TX beam—here, UL TX beam 381—for accessing both the ROs 415, 418. In a second option (option II), it would be possible that the UE 102 uses different UL TX beams—here, UL TX beams 381 and 384—for accessing the ROs 415, 418. It would also be possible that for some ROs different UL TX beams are used, while for other ROs the same UL TX beams are used.

In option I, the BS 101 can coherently combine signals received using multiple UL RX beams with optimized phase

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shift and/or amplitude, thereby obtaining a higher multi-antenna gain for rank-1 transmissions. The UL TX beam **381** used in option I to access, both, the RO **415**, as well as the RO **418** is determined in accordance with the DL TX beam **361**. For example, it would be possible that the UE **102** determines an angle of arrival of the SSB **411** transmitted using the DL TX beam **361**, e.g., based on a receive property of a corresponding RS included in the SSB **411**, e.g., the amplitude and/or phase at the various antennas **1024** of the UE **102**. Then, the UE **102** could determine the appropriate antenna weights so as to transmit UL signals with the same spatial characteristics, i.e., to invert the direction of the spatial data stream. This can be helpful under the assumption of channel reciprocity.

On the other hand, the UL TX beams **381**, **384** used in option II to access the RO **415** and the RO **418**, respectively, is determined in accordance with the DL TX beams **361** and **364**, respectively. The DL TX beam **361** is used to transmit the SSB **411** that is associated with the RO **415** which is accessed using the UL TX beam **381**. The DL TX beam **364** is used to transmit the SSB **411** that is associated with the RO **418** which is accessed using the UL TX beam **384**.

Thereby, according to option I, it is possible to configure a spatial diversity multi-antenna mode in which a single DL RX beam—i.e., DL RX beams **381*** corresponding to the UL TX beam **381**—is used at the UE **102** (cf. FIG. 5). The BS **101** can use the DL TX beams **361**, **364** to generate two dependent spatial data streams. The UE **102** can indicate, in a respective device parameter of the multi-beam operation that it prefers the diversity multi-antenna mode, and/or that it has coherent decoding/encoding capability, and/or whether the diversity multi-antenna mode is for UL transmission and/or DL transmission.

According to option II, spatial diversity or spatial multiplexing multi-antenna operational modes can be used, relying on the multiple DL RX beams **381*** and **384***. Multiple beam pairs between UE **102** and BS **101** can be established, already during initial access; these beam pairs can be used after initial access, when communicating payload data. The UE can indicate, in a respective device parameter of the multi-beam operation, that it prefers spatial-multiplexing or diversity multi-antenna mode, and/or has coherent or non-coherent operation, and/or whether the respective multi-antenna mode is for UL transmission and/or DL transmission.

FIG. 9 is a signaling diagram of communication between the BS **101** and the UE **102**. FIG. 9, in particular, illustrates aspects with respect to beam management during initial access, in accordance with FIG. 8. This is one illustrative example; other signal flows are possible.

FIG. 9 illustrates aspects with respect to communicating a configuration regarding access to multiple ROs. The configuration can be communicated from the communications NW **100** to the UE **102**—cf. NW configuration **6011**—and/or from the UE **102** to the communications NW **100**—cf. device configuration at **6506**. The multi-RO access then is in accordance with such configuration. For example, such configuration can impose certain constraints on the multi-RO access, e.g., in terms of the number of ROs accessed and/or in terms of the UL TX beams used by the UE for accessing the ROs. The multi-beam operation can be set in accordance with the configuration. Specifically, by communicating the configuration, it is possible for the communications NW **100** to appropriately configure the multi-beam operation. For instance, the UE **102** could indicate that it will use the same UL TX beam for accessing multiple ROs; then, the cellular NW can configure a diversity multi-antenna

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mode using multiple spatial streams that rely on a single UE beam. The rank of the multi-beam operation may be set in accordance with the indicated number of UL TX beams used by the UE for accessing the ROs.

Initially, at **6501**, the UE **102** receives a network configuration **6011** transmitted by the BS **101** of the cellular network **100**. The network configuration **6011** is indicative of a network support of the UE **102** accessing multiple ROs **415-418**. Thus, the access to more than a single RO **415-418** can be conditional on the respective network support as indicated by the network configuration **6011**. If the cellular NW **100** does not support multi-RO access, the UE **102** may only access a single RO **415-418** (e.g., except for escalation strategies where access on a given RO **415-418** fails).

There are various options available of providing the NW configuration to the UE **102**. For instance, the NW configuration **6011** could be received on the PDCCCH or as a control message on the PDSCH, e.g., as a RRC Layer 3 control message, while operating the UE **102** in the connected connection mode **401**. It would also be possible that the NW configuration **6011** is included in a system information block that is broadcasted by the BS **101**. In yet another option, it would be possible that the NW configuration **6011** is received during a RA procedure as early data **6012**, e.g., at **6507** (as will be explained in detail below).

It would be possible that the NW configuration **6011** is indicative of a maximum number of ROs **415-418** to be accessed by the UE **102**. For example, the NW configuration **6011** could be indicative of the UE **102** being allowed to access four or fewer ROs **415-418**. Then, the UE **102** may determine the appropriate number of ROs **415-418** to be accessed based on, e.g., its multi-beam operation configuration, a receive property of the RSs included in the SSBs **411** that are transmitted using the multiple beams **361-364**, etc., as well as in accordance with the maximum number of ROs.

The NW configuration **6011** can be indicative of a minimum number of UL TX beams to be used by the UE **102** when accessing the ROs **415-418**. For example, the NW configuration **6011** could be indicative of the number of UL TX beams expressed in terms of the number of ROs **415-418** to be used. For example, the NW configuration **6011** could be indicative of the UE **102** to use M UL TX beams when accessing N ROs **415-418**. M could be equal or smaller than N. M could be a function of N, i.e., M(N).

The NW configuration **6011** could also be indicative of a set number of UL TX beams to be used by the UE **102** when accessing the ROs **415-418**. I.e., the NW may set the number of UL TX beams to a certain value. The UE **102** may not have any choice regarding the selection of the number of UL TX beams to be used when accessing the ROs **415-418**.

Prior to initiating the RA procedure **600**, the UE **102** monitors for SSBs **411** transmitted by the BS **101**. This transmission is implemented using a beamsweep, e.g., the beamsweep **360** (cf. FIG. 8). The SSBs **411** transmitted at **6502** thus form a burst **410**. Based on the SSBs **411**, the UE **102** can synchronize with the BS **101**.

The UE **102** can then monitor for paging indicators and paging messages (paging signals) **6000**. Blind decoding of the PDCCCH for receiving the paging indicator can be implemented.

Upon receiving the paging signal **6000** at **6503**, initial access is triggered, in this example using RA procedure **600**.

As a general rule, there are other trigger criteria for initial access, e.g., UE power up or UE-originating UL data. As such, the paging signal **6000** is optional.

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At **6504**, based on the SSBs **411**, the UE **102** transmits a RA preamble to the BS **101**, in a respective RAmSg1 **6001**. Thereby, the UE **102** accesses RO **415**. This RAmSg1 **6001** may be indicative of a temporary identity of the UE **102** (UE-ID).

In response to transmitting the RAmSg1 **6001**, the UE **102** receives, at **6505**, a RA response message, the RAmSg2 **6002**. The RAmSg2 includes a new temporary identity for the UE **102**, timing adjustment information, and an UL scheduling grant for time-frequency resources. The UL scheduling grant may be addressed to the UEs **102** RA Radio NW Temporary Identity (RA-RNTI). Using these UL resources indicated by the UL scheduling grant included in the RAmSg2 **6002**, the UE **102** sends, at **6506**, a RRC connection request RAmSg3 **6003**. In response to the RRC connection request **6003**, the UE **102** receives, at **6507**, a contention resolution message RAmSg4 **6004** to ensure the right UE is addressed. This finalizes or aborts establishment of the data connection **189**. The UE **102** is then transitioned into operation into the connected connection mode **401**.

Early data (ED) **6012**—e.g., application data or RRC control data—can be included in the RAmSg3 **6003** and/or the RAmSg4 **6004**. Thus, generally UL ED and/or DL ED is possible to be included in the RA procedure. The ED **6012** can be piggybacked to the RAmSg3 **6003** and/or the RAmSg4 **6004**, e.g., in a NAS field. ED **6012** could also be included in RAmSg2, or even in RAmSg1, e.g., by using preamble partitioning. For a 2-step RA procedure, ED can be included in the RAmSgA and RAmSgB (corresponding to RAmSg1 and RAmSg3, and RAmSg2 and RAmSg4, respectively).

At **6508**, a further RA preamble **6001** is transmitted by the UE **102**, thereby accessing the RO **418**. As will be appreciated, the ROs **415** and **418** can be accessed based on the same measurements of the burst **410** of the SSBs **411** transmitted by the BS **101** at **6502** (in other examples, further bursts **410** of SSBs **411** may be monitored). Further, the RO **418** is accessed even though the access to the RO **415** has been successful, i.e., the multi-RO access is not due to an escalation strategy in case of failed RO access.

The UL ED **6012** can be used to provide a device configuration associated with the UE **102** accessing multiple ROs **415-418** to the cellular NW **100**. The UE **102** can access the multiple ROs **415-418** in accordance with the device configuration and the BS **101** can monitor the access of the ROs **415-418** in accordance with the device configuration.

For example, the device configuration could be indicative of the UE accessing multiple ROs. For instance, the device configuration could be indicative of the RO **418** that is later on accessed, i.e., of the to-be-accessed ROs. For example, the device configuration could be indicative of whether the same or different UL TX beams are used for accessing the multiple ROs **415-418** (cf. FIG. **8** where two options I and II have been explained, using the same UL TX beam **381** or using different UL TX beams **381**, **384**). It would be possible that the device configuration is indicative of the number of UL TX beams that are used by the UE **102** for accessing the multiple ROs **415-418**. Alternatively or additionally, the device configuration could be indicative of a number of ROs **415-418** that are accessed by the UE **102**.

While FIG. **9** illustrates an example in which the device configuration is transmitted in the RA procedure **600** associated with the RO **415**, in other examples, it would also be possible that the device configuration is provided otherwise to the cellular NW **100** (e.g., after transitioning to the connected connection mode **401**).

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The device configuration can be determined in accordance with the NW configuration **6011** received at **6501**. For example, the number of ROs **415-418** that are accessed by the UE **102** could be selected to be equal or smaller than the maximum number of ROs to be accessed as indicated by the NW configuration **6011**.

Alternatively or additionally to transmitting the device configuration of said accessing of the multiple ROs **415**, **418**, the ED **6012** can also be used to provide a device parameter of the multi-beam operation to the cellular NW **100**. Then, the multi-beam operation can later be established in accordance with the device parameter. The device parameter can be indicative of one or more UE capability limitations in connection with the multi-beam operation. For instance, the UE capability limitation could specify the inability of the UE **102** to coherently receive and/or transmit on multiple beams. The UE capability limitation could specify the inability of the UE **102** to communicate bidirectionally on some of the beams, e.g., single UL, single DL, bidirectional. For instance, the device parameter could indicate a preferred multi-beam operation mode such as diversity or spatial multiplexing or beamsweep. It would be possible that the device parameter includes an analog and/or digital front end capability of the interface **1022** of the UE **102**, e.g., whether the UE **102** supports digital beamforming, number of receive chains, a number of transmit chains, etc., to give just a few examples. Alternatively or additionally, it would be possible that the device parameter includes a beam priority order of the DL TX beams **361**, **364** used by the BS **101** to transmit the SSB **411** associated with the selected ROs **415**, **418** (cf. FIG. **8**). The beam priority order could be set, e.g., in accordance with a receive quality of the respective RSs included in the SSB **411**. For instance, the receive quality could be determined depending on a receive amplitude. To give an example, in a scenario in which a RS included in the SSB **411** transmitted using the DL TX beam **361** is received at the higher signal-to-noise level than a RS included in the SSB **411** transmitted using the DL TX beam **364**, the beam priority order could indicate that the DL TX beam **361** has priority over the DL TX beam **364**. This can be considered when establishing, e.g., spatial multiplexing multi-antenna operation so that data is routed at a lower data rate via the spatial data stream associated with the DL TX beam **364**, if compared to the spatial data stream associated with the DL TX beam **361**. The device parameter may be indicative of whether or not the UE **102** uses multiple beams for accessing the multiple ROs. The device parameter could be indicative of a number of beams that the UE **102** uses for accessing the multiple ROs.

While FIG. **9** illustrates an example in which the device parameter is transmitted in the RA procedure **600** associated with the RO **415**, in other examples, it would also be possible that the device parameter is provided otherwise to the cellular NW **100** (e.g., after transitioning to the connected connection mode **401**).

For example, the beam priority order as described above could also be implicitly indicated to the cellular NW **100** based on the sequence with which the ROs **415-418** are being accessed. For example, the scenario FIG. **8** and FIG. **9**, the RO **415** is accessed before accessing the RO **418**. This indicates the DL TX beam **361** having higher priority if compared to the DL TX beam **364**, because the DL TX beam **361** is used for transmitting the SSB **411** that is associated with the RO **415** and the DL TX beam **364** is used for transmitting the SSB **411** that is associated with the RO **418**.

In the example of FIG. **9**, the DL ED **6012** could be used in order to indicate which RA preamble **6001** is to be used

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by the UE 102 for accessing the RO 418 at 6508. Accordingly, the further RA preamble 6001 transmitted at 6508 can be contention free. The contention-free RA preamble could be provided as part of the DL ED 6012. Differently, the RA preamble 6001 transmitted at 6504 when accessing the RO 415 could be contention-based.

As a general rule, it would be possible that multiple ROs 415-418 are accessed with (i) contention-based random-access preambles only, (ii) contention free RA preambles only, or (iii) a mixture of contention-based and contention free RA preambles, as illustrated in FIG. 9. Details with respect to (i)-(iii) are explained in Table 1.

TABLE 1

Options for choice of RA preamble when accessing multiple ROs	
Choice of RA preamble	Explanation
(i)	Multiple contention-based preambles are used in multiple ROs. For example, different or the same contention-based preamble may be used. ED transmitted in the first RA procedure can include device configuration of multi-RO access and/or device parameter of multi-beam operation.
(ii)	Multiple contention-free preambles are used. NW configuration of multi-RO access, UE-ID can be provided prior to initial access to the UE 102.
(iii)	UE-ID acquired from first-accessed RO that is accessed using contention-based RA preamble. Subsequent ROs are accessed using contention-free RA preambles. This enables to identify the UE instantaneously and establish the multi-beam operation. NW configuration of multi-RO access can be provided as ED in RAmgs2 or RAmgs4 or RAmgsB. ED transmitted in the first RA procedure can include device configuration of multi-RO access and/or device parameter of multi-beam operation.

While FIG. 9 illustrates an example of a 4-step RA procedure 600, similar techniques may be readily employed for a 2-step RA procedure.

FIG. 10 is a flowchart of a method according to various examples. The method of FIG. 10 may be executed by a UE connectable to a cellular NW. For instance, the method of FIG. 10 could be executed by the UE 102. More specifically, the method of FIG. 10 could be executed by the processor 1021 of the UE 102, upon loading program code from the memory 1025.

Optional boxes are indicated with dashed lines in FIG. 10.

At box 3001, the UE receives a NW configuration from the cellular NW. The NW configuration is indicative of a NW support of the UE accessing multiple ROs for the purpose of configuring a multi-beam operation. In particular, the NW configuration can be received from a BS of a cell of the cellular NW that thereby indicates its support of the UE accessing multiple ROs. Alternatively or additionally, it would also be possible that the NW configuration is received from an anchoring node associated with the UE, e.g., a mobility control node such as the AMF 131 (cf. FIG. 6).

Next, at box 3002, the UE checks whether it is to perform an initial access. If yes, the method proceeds to box 3003.

At box 3003, the UE monitors for SSBs. The UE receives one or more SSBs that can have different time indices. Different SSBs within a burst are transmitted using different DL TX beams. The SSBs include DL RSs. Each SSB is associated with a respective ROs.

The UE can then select, at box 3004, at least two of these ROs. The selected at least two ROs are associated with at least two DL RSs included in the received SSBs that are selected by the UE based on the monitoring at box 3003. For

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instance, the N strongest DL RSs (i.e., being received with the highest signal-to-noise ratio) could be selected.

At box 3005, the selected at least two ROs are accessed. I.e., RA preambles are transmitted at the selected at least two ROs. Contention free and/or contention-based RA preambles may be used.

At box 3006, it is optionally possible to transmit a device configuration of said accessing of the at least two ROs and/or a device parameter for the subsequent multi-beam operation to the cellular NW. For instance, this can be done using ED piggybacked to the RAmgs3 or RAmgsA.

Note that box 3006 can be executed before the access to the multiple ROs at box 3005 is completed.

At box 3007, multi-beam operation of a wireless link between the UE and the cellular NW is established. The multi-beam operation is established in accordance with the access to the multiple ROs at box 3005. For example, it would be possible that multiple DL TX beams are used for the multi-beam operation, wherein the multiple DL TX beams are associated with the ROs that are accessed at box 3005. More specifically, the multiple DL TX beams that are used for the multi-beam operation that is established at box 3007 can be used for transmitting the SSBs associated with the selected at least two ROs of box 3004. It would also be possible to refine these beams, e.g., use sub-beams.

The multi-beam operation may be used after the RA procedure has concluded.

For example, a selection could be made between a diversity multi-antenna mode or a spatial multiplexing multi-antenna operation mode, in accordance with the access to the multiple ROs at box 3005.

At box 3008, it is then possible to establish the wireless link between the UE and the cellular NW, e.g., by completing a RA procedure triggered by the access to one of the ROs that are accessed at box 3005. The RA procedure can be completed using the multi-beam operation that is established at box 3007. Further, subsequent communication on the then established wireless link can be implemented using the multi-beam operation that is established at box 3007.

The order of boxes 3001-3008 is an example only. For example, it would be possible that the NW configuration is received as part of a RA procedure that is triggered by the RO access of box 3005 (as such, box 3001 can be executed after box 3005).

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FIG. 11 is a flowchart of a method according to various examples. The method of FIG. 11 can be executed by an access node of a communications NW. For instance, the method of FIG. 11 could be executed by a BS of a cellular NW, e.g., the BS 101 of the cellular NW 100. More specifically, it would be possible that the method of FIG. 11 is executed by the processor 1011 of the BS 101, upon loading program code from the memory 1015.

The method of FIG. 11 is interrelated with the method of FIG. 10. I.e., the BS 101 and the UE 102 interact when executing the methods of FIG. 10 and FIG. 11.

Optional boxes are illustrated with dashed lines in FIG. 11.

The method starts with box 3101. Here, a NW configuration is transmitted to a UE that is in communication with the access node. Box 3101 is interrelated with box 3001. The NW configuration can indicate whether the access node supports the UE accessing multiple ROs, in order to establish a multi-beam operation during initial access. Box 3101 is interrelated with box 3001.

At box 3103, SSBs are transmitted using multiple beams. In particular, a burst of SSBs can be transmitted using a beam sweep. The SSBs include RSs. The UE can sound the channel based on the RSs. Each SSB is associated with a respective RO. Box 3103 is interrelated with box 3003.

At box 3105, the ROs are monitored. I.e., the access node attempts to receive a RA message on the ROs. For example, a BS of a cellular NW can attempt to receive RAmSg1 or RAmSgA on the ROs. Box 3105 is interrelated with box 3005.

By receiving RAmSg1 or RAmSgA on multiple ROs the BS can determine that the same UE has accessed both RAs.

In some examples, it would be possible that based on a receive property—i.e., amplitude and/or phase—of a RS associated with the RA message the BS acquires channel state information. The subsequent multi-beam operation can be configured based on such channel sounding.

At box 3106, it would be possible to receive a device configuration for the multi-RO access of the UE and/or a device parameter of the subsequent multi-beam operation. Box 3106 is interrelated with box 3006.

Then, at box 3107, the multi-beam operation is established. For instance, a diversity multi-antenna mode or a spatial multiplexing multi-antenna operation may be activated. The beams used for the multi-beam operation can be determined based on the ROs that are accessed by the UE at box 3105. For instance, each RO can be associated with the respective DL TX beam that has been used to transmit the corresponding SSB. Then, these DL TX beams can be used for the multi-beam operation, or be further refined. For example, the multi-beam operation pertains to UL transmission, then UL RX beams can be used that are corresponding to these DL TX beams, e.g., use the same antenna weights and/or having corresponding spatial characteristics.

The multi-beam operation that is established at box 3107 can be used for completing RA on further accessed ROs, as part of box 3105. The multi-beam operation can also be used for communicating data on the wireless link that is subsequently established at box 3108. Box 3107 corresponds to box 3007. Box 3108 corresponds to box 3008.

Summarizing, above techniques have been described that facilitate multi-beam operation, e.g., using diversity multi-antenna and/or spatial multiplexing multi-antenna operation. Techniques have been described that facilitate a UE accessing multiple ROs. The cellular NW can indicate whether it

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support such multi-RO access. For example, such NW configuration can be signaled as part of a broadcasted system information.

The UE can access a first RO and report its intention to access one or more further ROs, e.g., using ED transmitted as part of the RA procedure associated with the access of the first RO. as a general rule, the UE can provide a device configuration for the multi-RO access and/or a device parameter for the multi-beam operation to the cellular NW, e.g., as ED. The UE could indicate the expected number of ROs to be accessed and/or the number of UL TX beams used for this. For instance, the UE may use the same UL TX beam for accessing two or more of multiple ROs.

The BS may assign RA preambles for the one or more further ROs as part of DL ED transmitted to the UE as part of the RA procedure associated with the access of the first random access occasion. Then, contention-free access is possible in the at least one further RA procedure associated with the one or more further ROs.

By means of such techniques, low delay to spatial diversity and/or multiplexing is possible already during initial access. The overall control signaling overhead can be limited.

Although the invention has been shown and described with respect to certain preferred embodiments, equivalents and modifications will occur to others skilled in the art upon the reading and understanding of the specification. The present invention includes all such equivalents and modifications and is limited only by the scope of the appended claims.

For illustration, above, various techniques have been described in which a 4-step RA procedure is used. Similar techniques may also be applied for a 2-step RA procedure. As a general rule, the 2-step RA procedure relies on a combination of RAmSg1 and RAmSg3 into RAmSgA, and on a combination of RAmSg2 and RAmSg4 into RAmSgB.

For further illustration, various examples have been described in which UL ED is piggybacked to RAmSg3. It is possible that at least parts of this information is provided via the RAmSg1, e.g., using preamble partitioning.

For further illustration, various examples have been described in which a DL RS is implemented by SSB. In other examples, other kinds and types of DL RS may be used.

For still further illustration, various examples have been described in which the UE selects the strongest DL TX beams. In other examples, other and/or further quality metrics could be taken into account, e.g., signal-to-noise ratio.

For still further illustration, above, various scenarios have been described in which a single BS receives RAmSg1 or RAmSgA on multiple ROs accessed by a UE and monitored by that single BS. Similar techniques may be applicable to scenarios in which multiple ROs associated with SSBs of a burst and accessed by a UE are monitored by different BSs, e.g., through TRPs.

The invention claimed is:

1. A method of operating a wireless communication device configured to connect to a communications network, the method comprising:

monitoring for multiple downlink reference signals transmitted by at least one access node of the communications network using multiple downlink transmit beams, each one of the multiple downlink reference signals being associated with a respective at least one of multiple random-access occasions,

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accessing at least two random-access occasions of the multiple random-access occasions, the at least two random-access occasions being associated with at least two downlink reference signals selected from the multiple downlink reference signals based on said monitoring, and

establishing a multi-beam operation between the wireless communication device and the communications network, wherein the multi-beam operation is established based on at least two downlink transmit beams used for transmitting the selected at least two downlink reference signals.

2. The method of claim 1, wherein said accessing of the at least two random-access occasions is in accordance with at least one configuration communicated between the wireless communication device and the communications network.

3. The method of claim 1, further comprising: receiving a network configuration from the communications network, wherein the at least two random-access occasions are accessed in accordance with the network configuration.

4. The method of claim 3, wherein the network configuration is indicative of a maximum number of random-access occasions to be accessed by the wireless communication device.

5. The method of claim 3, wherein the network configuration is indicative of whether the wireless communication device is to use the same or different uplink transmit beams are used for said accessing of the at least two random-access occasions.

6. The method of claim 3, wherein the network configuration is indicative of a minimum number of uplink transmit beams to be used by the wireless communication device for said accessing of the at least two random-access occasions, wherein the minimum number of uplink transmit beams optionally depends on a number of the at least two random-access occasions.

7. The method of claim 3, wherein the network configuration is indicative of a network support of access to multiple random-access occasions.

8. The method of claim 1, further comprising: transmitting a device configuration of said accessing of the at least two random-access occasions to the communications network, wherein said accessing of the at least two random-access occasions is in accordance with the device configuration.

9. The method of claim 8, wherein the device configuration is indicative of whether the same or different uplink transmit beams are used for said accessing of the at least two random-access occasions.

10. The method of claim 8, wherein the device configuration is indicative of a number of random-access occasions accessed by the wireless communication device.

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11. The method of claim 8, wherein the device configuration is transmitted in a random-access procedure associated with at least one of the at least two random-access occasions.

12. The method of claim 8, wherein the device configuration is indicative of a number of uplink transmit beams that are used for said accessing of the at least two random-access occasions.

13. The method of claim 8, further comprising: receiving a network configuration from the communications network, and wherein the at least two random-access occasions are accessed in accordance with the network configuration; and wherein the device configuration is determined based on the network configuration.

14. The method of claim 1, further comprising: transmitting a device parameter of the multi-beam operation to the communications network, wherein the multi-beam operation is established in accordance with the device parameter.

15. The method of claim 14, wherein the device parameter comprises a mode of the multi-beam operation selected from: diversity, and spatial multiplexing.

16. The method of claim 14, wherein the device parameter comprises a frontend capability of a wireless interface of the wireless communication device.

17. The method of claim 14, wherein the device parameter comprises a beam priority order of the at least two downlink transmit beams, the beam priority order depending on a receive property of the at least two downlink reference signals.

18. The method of claim 14, wherein the device parameter is transmitted in a random-access procedure associated with said accessing of the at least two random-access occasions.

19. The method of claim 1, wherein the at least two random-access occasions are accessed using the same uplink transmit beam.

20. A method of operating at least one access node of a communications network, the method comprising: transmitting multiple downlink reference signals using multiple downlink transmit beams, each one of the multiple downlink reference signals being associated with a respective at least one of multiple random-access occasions, monitoring an access of the wireless communication device to at least two random-access occasions associated with at least two downlink reference signals of the multiple downlink reference signals, and based on said monitoring of the access, establishing a multi-beam operation between the wireless communication device and the communications network, wherein the multi-beam operation is established based on at least two downlink transmit beams of the multiple downlink transmit beams used for transmitting the at least two downlink reference signals.

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